DR. PRIYA N. RAGHUNATHAN

ASSOCIATE PROFESSOR OF COMPUTATIONAL BIOLOGY

p.raghunathan@example.edu , +1 (412) 555-0187 , Pittsburgh, PA , raghunathanlab.example.edu , github.com/raghunathan-lab-example

RESEARCH INTERESTS

Statistical and machine-learning methods for single-cell and spatial transcriptomics; probabilistic models of gene regulatory networks; reproducible computational pipelines for clinical genomics.

ACADEMIC APPOINTMENTS

Associate Professor (with tenure) Jul 2021 – Present
Carnegie Mellon University, Computational Biology Department • Pittsburgh, PA

- Lead a group of 7 Ph.D. students, 2 postdocs and 3 research staff.
- Courtesy appointment in Machine Learning Department (2022 - present).

Assistant Professor Jul 2016 – Jun 2021
Carnegie Mellon University, Computational Biology Department • Pittsburgh, PA

- Established the lab; graduated 3 Ph.D. students.

Postdoctoral Fellow Sep 2013 – Jun 2016
Broad Institute of MIT and Harvard • Cambridge, MA

- Single-cell methods group; Jane Coffin Childs Fellow.

JOURNAL ARTICLES

Spatially resolved gene regulatory networks from sparse transcriptomic maps 2025
Raghunathan P.N.*, Okoye T., Brandt L., Feld S. Nature Methods 22(4):511-523

Calibrated uncertainty for cell-type deconvolution 2024
Okoye T., Liang Y., Raghunathan P.N. Genome Biology 25:188

scNoise: a hierarchical model of transcriptional bursting at single-cell resolution 2023
Liang Y., Raghunathan P.N. Cell Systems 14(2):97-110

Benchmarking spatial clustering methods across 42 tissue atlases 2022
Brandt L., Herrera A., ..., Raghunathan P.N. Nature Communications 13:6120

Reproducible clinical genomics pipelines with provenance tracking 2021
Feld S., Raghunathan P.N. Bioinformatics 37(19):3311-3318

Variational inference for allele-specific expression in tumors 2020
Herrera A., Kim J., Raghunathan P.N. PLOS Computational Biology 16(8):e1008021

EDUCATION

University of California, Berkeley | 2008 – 2013
Ph.D. in Computational Biology
Dissertation: Hierarchical Bayesian models of transcriptional noise. Advisor: Prof. M. Lindqvist.

University of Michigan | 2006 – 2008
M.S. in Statistics

Indian Institute of Technology Madras | 2002 – 2006
B.Tech. in Biotechnology

SKILLS

R · Python · Stan · PyTorch · Nextflow · Slurm/HPC · scRNA-seq · Spatial transcriptomics

LANGUAGES

English • Tamil • Hindi

AWARDS

NSF CAREER Award (2019)

Sloan Research Fellowship in Computational & Evolutionary Molecular Biology (2018)

CMU School of Computer Science Teaching Award (2020)

ISMB Outstanding Paper Award (2017)

Jane Coffin Childs Postdoctoral Fellowship (2013 - 2016)

LINKS

github.com/raghunathan-lab-example
raghunathanlab.example.edu

Gene-gene covariance from droplet-based scRNA-seq 2019
Raghunathan P.N., Lindqvist M. Annals of Applied Statistics 13(3):1601-1630

Single-cell atlas of the developing human thymus 2018
Thymus Atlas Consortium (incl. Raghunathan P.N.). Science 367(6480):eaay3224

Noise-aware differential expression for low-input RNA-seq 2017
Raghunathan P.N., Weiss D., Regev-Haddad K. Genome Research 27(9):1525-1536

Hierarchical Bayesian models of transcriptional noise 2015
Raghunathan P.N., Lindqvist M. Journal of the American Statistical Association 110(512):1406-1419

CONFERENCE PROCEEDINGS

Okoye T., Raghunathan P.N. Optimal transport for spatial alignment of serial sections. ICML 2024, pp. 38112-38129., Liang Y., Okoye T., Raghunathan P.N. Contrastive pretraining for gene programs. RECOMB 2023, LNCS 13976:201-216., Brandt L., Raghunathan P.N. Graph neural networks for tissue niches. ISMB 2022 (Bioinformatics 38 Suppl 1)., Herrera A., Raghunathan P.N. Amortized inference for clonal deconvolution. NeurIPS 2020, pp. 11843-11855., Raghunathan P.N. Sparse factor models for single-cell multi-omics. ISMB 2017 (Outstanding Paper Award)., Raghunathan P.N., Weiss D. Bursting kinetics from snapshot data. RECOMB 2016, LNCS 9649:254-268.

GRANTS & FUNDING

R01 GM148812: Probabilistic models of spatial gene regulation 2023 – 2028
National Institutes of Health (NIGMS). PI. $2,450,000

CAREER: Uncertainty-aware machine learning for single-cell genomics 2019 – 2025
National Science Foundation (DBI-1945xxx). PI. $812,000

Chan Zuckerberg Initiative Single-Cell Biology Data Insights 2022 – 2024
CZI. Co-PI (with S. Feld). $400,000

Sloan Research Fellowship 2018 – 2020
Alfred P. Sloan Foundation. PI. $75,000

Pilot: Clinical-grade pipelines for pediatric tumor sequencing 2017 – 2018
UPMC Hillman Cancer Center. Co-I. $150,000

TEACHING

02-710 Computational Genomics (graduate) 2017 – Present
Carnegie Mellon University. Instructor, ~60 students/yr. FCE 4.7/5

02-251 Great Ideas in Computational Biology (undergraduate) 2019 – Present
Carnegie Mellon University. Instructor, ~120 students/yr

02-750 Automation of Scientific Research (graduate seminar) 2021 – 2023
Carnegie Mellon University. Co-instructor

Workshop: Bayesian Methods for Single-Cell Data 2018, 2022
Cold Spring Harbor Laboratory course. Lecturer

INVITED TALKS

Keynote: Uncertainty is a feature, not a bug, in spatial omics 2025
Single Cell Genomics Conference, Stockholm

Spatial gene regulation from sparse maps 2024
Department Seminar, Stanford University

Calibrated deconvolution for clinical samples 2024
Broad Institute Models, Inference & Algorithms

Probabilistic programming for genomics 2023
Gordon Research Conference: Quantitative Genetics

Transcriptional bursting at scale 2022
EMBL Heidelberg Seminar Series

Benchmarks that matter 2022
Human Cell Atlas General Meeting

Noise models for single-cell data 2019
Joint Statistical Meetings, invited session

Reproducibility in computational biology 2018
NIH Data Science Symposium

PROFESSIONAL SERVICE

Associate Editor, PLOS Computational Biology (2021 - present), Program Committee, ISMB (2019 - 2025), RECOMB (2021 - 2024), NeurIPS (2020 - 2023), NIH Study Section ad hoc reviewer, GCAT and MABS (2020 - 2024), Reviewer: Nature Methods, Genome Biology, Cell Systems, Bioinformatics, JASA, Chair, CMU Computational Biology Ph.D. Admissions (2022 - 2024), Co-organizer, Pittsburgh Single-Cell Symposium (2019, 2023)